The AG Share of New Company Registrations in Basel-Landschaft, 2017 vs 2025

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Master of Science in AI, Capstone Project 2: Data and Statistical Reasoning

August 17, 2026

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Overview

In this project I analyzed 11,267 new company registrations that were published for the canton of Basel-Landschaft between 2017 and 2025, implemented in a Jupyter notebook (analysis.ipynb) with Python, NumPy, Pandas, Matplotlib, Seaborn and SciPy. I wanted to know whether the share of stock corporations (Aktiengesellschaft, AG) among all new company registrations changed between 2017 and 2025. The data comes from the official dataset Firmenmutationen nach Rechtsform und Gemeinde (seit Februar 2016), published by Open Government Data Kanton Basel-Landschaft at https://data.bl.ch/explore/dataset/12460/. The question follows up on my first capstone project, where the AG / SA was the largest legal form group in the Swiss LEI register, but the two projects use different datasets and different populations, so their percentages cannot be compared directly.

Dataset Description

The dataset contains every mutation message that was published for companies in the canton of Basel-Landschaft in the Swiss Official Gazette of Commerce (SOGC, commonly known by its German abbreviation SHAB) since February 2016 (Open Government Data Kanton Basel-Landschaft, 2026). The frozen copy I downloaded on August 13, 2026 has 28,992 rows and 11 columns, and one row is one single register message, so the data is not aggregated. The key variables for my question are the mutation category, the publication date and the legal form. I only used the 13,124 messages with the category Neueintragung (new registration), and inside this subset the official company identifier (UID) is unique, so one row is one newly registered company. I restricted the analysis to the complete calendar years 2017 to 2025 because the data starts in February 2016 and the year 2026 was still running at download time. After both filters 11,267 new registrations remain.

Methods

Before any inference I ran a structured screening of the data. I checked missing values, duplicate rows, the uniqueness of the message id and of the company identifier, the exact legal form labels and the date range. Lusa et al. (2024) recommend this kind of initial data analysis before the actual analysis because it can surface data properties that would otherwise silently change the interpretation of the results, and they also recommend fixing the analysis plan in advance to avoid ad hoc decisions. I followed both points, the hypotheses, the two comparison years and the AG definition were fixed before I computed any test result. The screening confirmed two documented quality notes of the publisher, the municipality columns are partly incomplete and the free text field contains encoding errors, and neither field is needed for my question.

For the descriptive part I report counts and shares per year and per legal form, because the mutation category and the legal form are categorical variables and counts and shares describe them directly. I added the mean and the standard deviation of the yearly registration counts to summarize the typical yearly level and the year-to-year variation. The endpoint counts and Figure 1 describe the growth of the total volume.

I built three visual models of the dataset, plus a fourth figure that visualizes the hypothesis test. A bar chart of the yearly volume (Figure 1) shows the growing base, a grouped bar chart (Figure 2) compares the legal form composition of 2017 and 2025, a line chart (Figure 3) tracks the AG share year by year, and a histogram (Figure 4) shows the simulated null distribution of the test. Figure 3 supports the research question most directly because it plots the quantity from the question itself, the other figures provide the context around it.

For the hypothesis test I compared the AG share of 2017 with the AG share of 2025, with a two-sided alternative and a significance level of alpha = 0.05. I compared the earliest and the latest complete year because the research question concerns the change across the full available window, and Figure 3 keeps every year in between visible, so the endpoint result is not presented as a smooth trend. The test is a parametric Monte Carlo test for a difference in proportions. Instead of an analytical formula I simulated the null distribution. Under the null hypothesis both years draw their AG registrations with the same probability, so I pooled the two years to estimate this probability, drew 100,000 binomial samples for each year with a fixed random seed and recorded the difference of the simulated shares. Hesterberg (2015) argues that resampling and simulation methods make the sampling distribution concrete and let the analyst focus on the idea of the test instead of a formula, and that is why I chose this approach, the p-value can be read directly from the simulated distribution. Hope (1968) shows that a valid significance test can be based on a finite set of simulated samples and that the power of such a test increases with the size of that set, which is why I drew 100,000 samples rather than a smaller reference set. Following that finite construction, the observed difference counts as one member of the reference set, so I compute the p-value as (r + 1) / (B + 1), where r is the number of simulated differences at least as extreme as the observed one and B = 100,000 is the number of draws. An analytical 2 x 2 chi-square test would test the same null hypothesis. I chose the simulation because it expresses the test statistic directly as the difference in AG shares and makes the null distribution visible in Figure 4. The smallest expected cell count under the null hypothesis is 164.06, which is far above the minimum expected count of 5 that Campbell (2007) reports as the common recommendation for the chi-square test of a 2 x 2 table, so the approximation would also be appropriate here. I report the cross-check without the continuity correction because my simulation draws the counts directly and applies no correction either. The simulation was the planned primary analysis, and the Pearson chi-square test serves only as an analytical cross-check of the same 2 x 2 table and the same null hypothesis, not as a second independent hypothesis test. The test treats the yearly totals as fixed, and with 1,156 and 1,444 registrations both comparison years are large. It also rests on two simplifying assumptions. It treats the registrations as independent of each other, although founding decisions can cluster in reality, and it treats the two years as draws from an underlying registration process, because the data is a complete count of both years and not a random sample. The p-value therefore describes how unusual the observed difference would be if that process had stayed the same, and I discuss both assumptions in the limitations section.

Results

Between 2017 and 2025 the yearly number of new registrations grew from 1,156 to 1,444 (Figure 1). Across the nine complete years the yearly registration count was M = 1,251.9 (SD = 111.8), and it ranged from 1,106 in 2018 to 1,444 in 2025. Two legal forms dominate throughout. The GmbH share stayed nearly unchanged, 39.9% in 2017 and 40.4% in 2025, while the sole proprietorship share rose from 36.1% to 39.5% (Figure 2). The AG moved in the opposite direction, its share fell from 16.00% (185 of 1,156) in 2017 to 12.74% (184 of 1,444) in 2025, a difference of -3.26 percentage points (Figure 3). The absolute number of AG registrations barely moved, so the falling share is driven by the growth of the other legal forms and not by fewer AG foundations.

Figure 1

New Company Registrations per Year in the Canton of Basel-Landschaft, 2017 to 2025

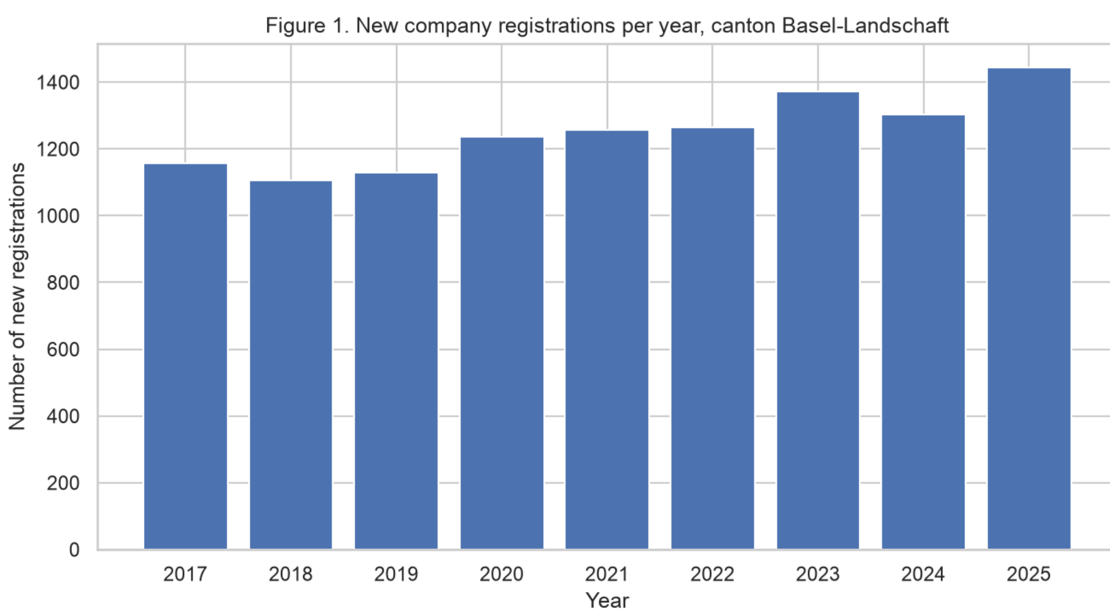
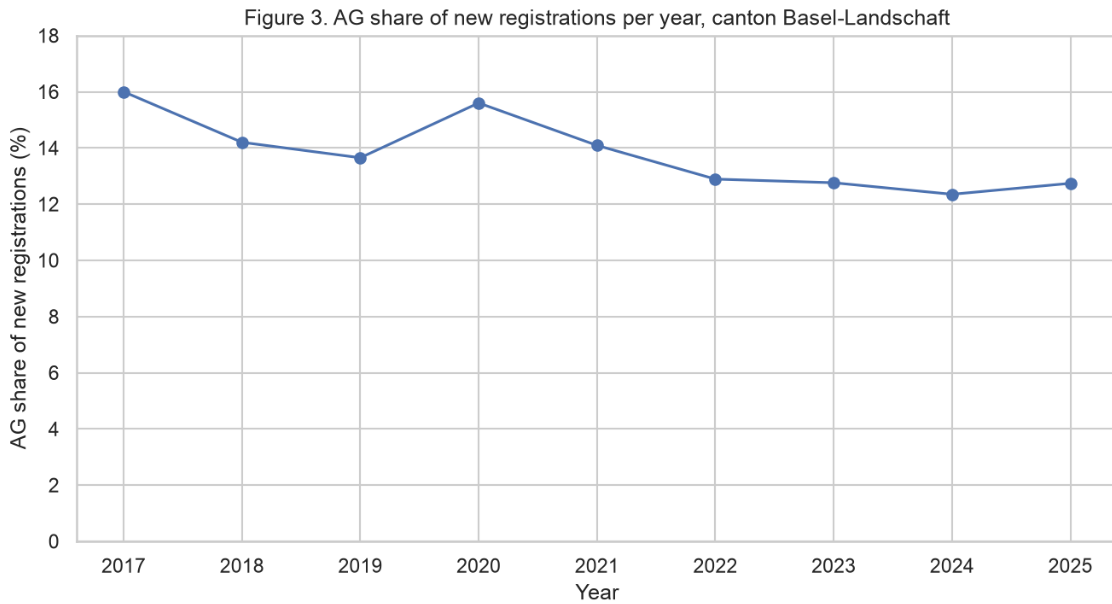


Figure 2

Legal Form Shares of New Registrations, 2017 vs 2025

Figure 3

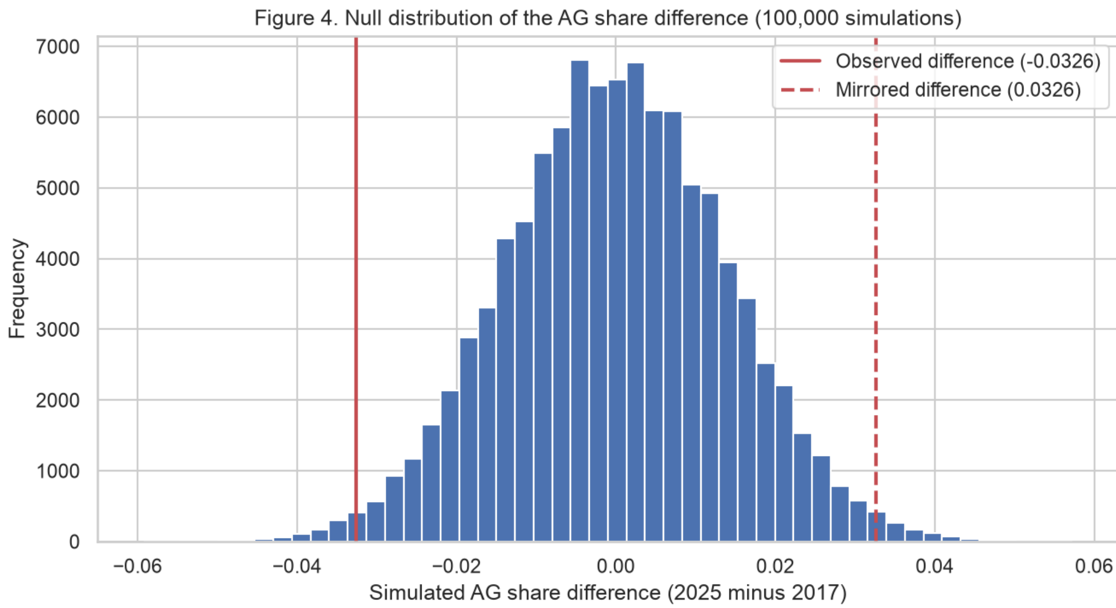
AG Share of New Registrations per Year, 2017 to 2025



The simulated null distribution is centered near zero with a standard deviation of 0.0138 (Figure 4). The observed difference of -0.0326 lies far in the left tail, and 1,820 of the 100,000 simulated differences (1.82%) were at least as extreme in either direction. The two-sided p-value of 0.0182 is below alpha = 0.05, so I reject the null hypothesis of an equal AG share in 2017 and 2025. As an analytical cross-check, the Pearson chi-square test without continuity correction gave chi-square(1) = 5.61 and p = 0.0179, so it supports the same decision as the planned simulation test.

Figure 4

Simulated Null Distribution of the AG Share Difference With the Observed Difference Marked



Interpretation for a Non-Technical Audience

Out of every 100 companies that were newly registered in Basel-Landschaft in 2017, 16 chose the legal form of a stock corporation. In 2025 it was fewer than 13 out of 100. At the same time more companies were founded overall, 1,444 in 2025 against 1,156 in 2017. The number of new stock corporations itself stayed almost the same, 185 in 2017 and 184 in 2025, so the stock corporation did not disappear, other legal forms simply grew around it.

A statistical test says that a gap of this size would rarely appear by chance alone if the registration behavior behind both years had stayed the same. In that case a gap this large would show up in fewer than 2 of 100 comparisons like this one. The calculation treats every registration as an independent decision, which is a simplification. The data also does not say why the share fell, it only shows that it fell, and it says nothing about how successful or stable these companies are.

Limitations and Potential Bias

The clearest limitation is the regional scope. The data covers one canton, and Basel-Landschaft is not a miniature of Switzerland, so the results must not be transferred to the whole country without checking other cantons. A second limitation is what the data represents. These are commercial register messages, not economic activity as a whole. The duty to register a sole proprietorship only starts at CHF 100,000 of yearly revenue (State Secretariat for Economic Affairs [SECO], n.d.), so the smallest businesses stay invisible, and the legal form of a registration says nothing about revenue, success or financial stability. A third limitation is the time frame, the dataset starts in February 2016 and the current year is unfinished, so I excluded both border years, and the online dataset is updated continuously, which is why my notebook works on a frozen copy that is committed to the repository.

The register data is also administrative data. Connelly et al. (2016) point out that such data is not collected for research purposes, so its coverage and content follow legal rules and not research needs, and that it typically records a complete population rather than a sample. Both points apply here. My test therefore treats the two years as draws from an underlying registration process, and it assumes independent registrations even though founding decisions can cluster, for example when one person registers several companies at once or when an economic event pushes many foundings into the same year. If registrations cluster, the simulated uncertainty is understated, so the p-value should be read as a guide and not as an exact error rate.

The comparison is observational, so the difference between 2017 and 2025 could come from economic conditions, legal changes or shifts in the mix of founded businesses, and my analysis cannot separate these causes. This creates a real interpretation bias risk. A reader could turn the falling share into the claim that the AG became unpopular, but the data only supports the weaker statement that the observed AG share among new registrations was lower in 2025 than in 2017, and the almost constant absolute AG numbers show how misleading the stronger claim would be. There is also a selection effect in the data itself, the register only shows activity that must be registered, so the dataset over-represents formal business activity. Finally I see an ethical point in this kind of data. Connelly et al. (2016) note that the public may have concerns about privacy and about the linkage of administrative records from different sources, and the free text field of my dataset contains the names of real persons. Analyses that link these names across registers to profile individuals would go far beyond the purpose of this public dataset, and I did not use this field in the analysis. A useful next step would be to repeat the same comparison for additional cantons and for later complete years to see whether the pattern generalizes beyond Basel-Landschaft.

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